

SUPERVISED MACHINE LEARNING - LOGISTIC REGRESSION (FRAUD)

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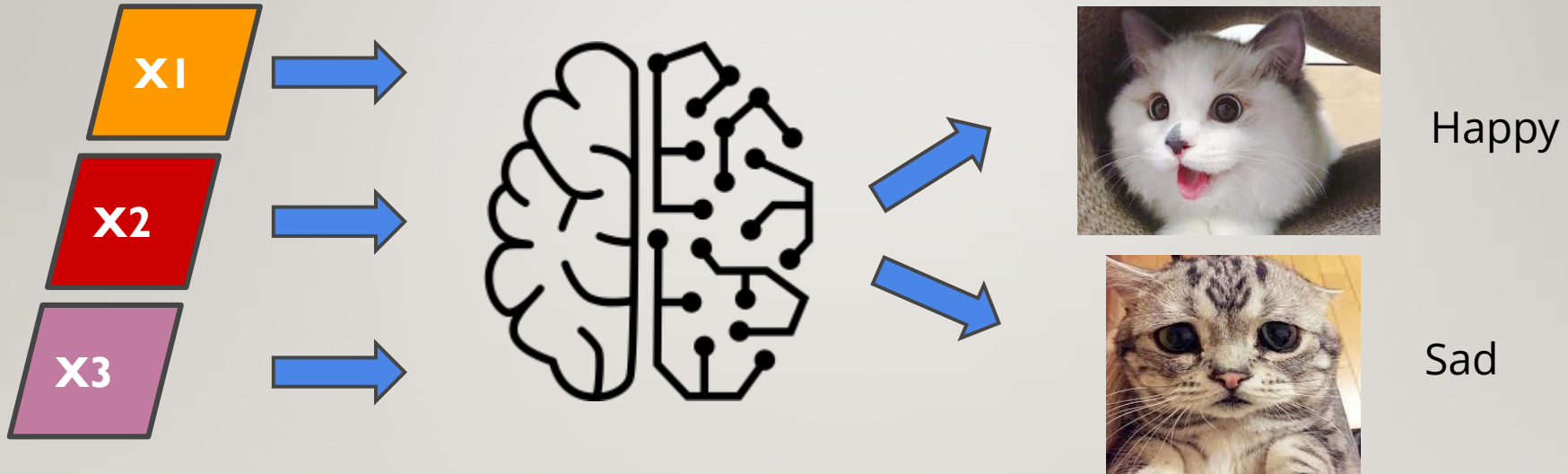
MACHINE LEARNING



Machine learning is a method devoted to learn from historical data then leverage it to create model that can be used to predict or help decision making.

Machine learning have many forms, one of the form that will used on this case is supervised machine learning based on logistic regression.

LOGISTIC REGRESSION



Logistic Regression is a model with binary output that based on several data input. This data input used to determine the output.

STUDY CASE

On this study case the example will be logistic regression for payment fraud detection. Data input will be like this table

accountAgeDays	numItems	localTime	paymentMethod	paymentMethodAgeDays	label
29	1	4.745402	paypal	28.204861	0
725	1	4.742303	storecredit	0.000000	0
845	1	4.921318	creditcard	0.000000	0
503	1	4.886641	creditcard	0.000000	0
2000	1	5.040929	creditcard	0.000000	0

RESULT - EXCLUDE PAYMENT METHOD

	accountAgeDays	numItems	localTime	paymentMethodAgeDays	label
0	29	1	4.745402	28.204861	0
1	725	1	4.742303	0.000000	0

accuracy = 1.0

Confussion Metrix

```
[[12753  0]  
[  0 190]]
```

On the result, the logistic model have accuracy 100% and based on confusion metrix 190 data are correctly indicate as fraud where 12753 not indicted as fraud.

RESULT - CHANGE PAYMENT METHOD VALUE

accountAgeDays	numItems	localTime	paymentMethod	paymentMethodAgeDays	label
29	1	4.745402	1	28.204861	0
725	1	4.742303	2	0.000000	0
845	1	4.921318	3	0.000000	0

accuracy = 1.0

Confussion Metrix

```
[[12753  0]
 [  0 190]]
```

On the result, the logistic model have accuracy 100% and based on confusion metrix 190 data are correctly indicate as fraud where 12753 not indicted as fraud.

THANK YOU!